

Information Gain Prefers the Questions Users Cannot Answer: Reliability-Aware Query Selection for Interactive Identification

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Abstract—Interactive identification systems narrow a catalogue to a single item by asking a sequence of binary questions, and the standard criterion for choosing the next question is expected information gain. We show that this criterion carries a systematic and previously unremarked defect when the answers come from a human. Information gain rewards questions that partition the belief evenly, but in attribute-indexed catalogues the attributes that partition most evenly are the vague, subjective ones—mood, energy, perceived danceability—which are precisely the attributes a user is least able to answer correctly. The bias is aggravated by a routine preprocessing step: binning a continuous feature at its median guarantees a maximally balanced split, and therefore maximal apparent informativeness, for exactly the features whose semantics are least crisp. We make three contributions. First, we formalise interactive identification with unreliable responses as sequential search over a per-attribute binary symmetric channel, and observe that hard candidate elimination is the zero-noise limit of Bayesian updating rather than a competing design. Second, we derive reliability-aware information gain, the mutual information between the target and the *observed* answer, and establish that it re-ranks candidate questions rather than merely rescaling them, that it is bounded by the per-attribute channel capacity, and that this bound yields a query-budget lower bound of $\log_2 N / \max_f (1 - H_b(\epsilon_f))$. Third, we show the modification is deployable without measuring user error rates: a three-tier annotation distinguishing objective, semi-objective and subjective attributes is sufficient to instantiate it. Against a real 79,803-track catalogue, reliability-aware selection (RAIG-tiered) raises top-1 accuracy from 2.1% to 7.9% over the strongest current-practice baseline at matched heterogeneous noise ($n = 900$ paired trials, exact McNemar $p = 1.8 \times 10^{-9}$), recovers the entire oracle-to-uniform accuracy gap from a three-tier annotation alone, and survives a lightweight correlated-error stress test. It reverses, exactly as predicted, when the reliability assumption is randomised or adversarially inverted – a failure mode we verify holds in 8 of 8 seeds under adversarial noise.

Index Terms—interactive identification, question selection, information gain, noisy search, Bayesian belief update, binary symmetric channel, human–computer interaction

I. INTRODUCTION

A. Motivation

A recurring interaction pattern places a machine in the role of questioner and a person in the role of oracle. A troubleshooting assistant narrows a fault by asking about

symptoms; a clinical triage tool narrows a differential by asking about presentation; a product finder narrows a catalogue by asking about requirements; the parlour game Akinator narrows a space of characters by asking about traits. In each case the system holds a belief over a discrete set of hypotheses, selects a question, observes an answer, and revises. The engineering question is which question to ask, and the near-universal answer is the one that maximises expected information gain.

That criterion inherits its authority from a clean theoretical setting in which the answer is observed exactly. Human answers are not observed exactly. A user asked whether the song they have in mind is “energetic” may hesitate, may apply a personal threshold, or may simply be wrong, and the system has no way to distinguish a considered answer from a guess. What makes this more than a routine robustness concern is that the unreliability is not spread uniformly over the question space. Some attributes admit near-certain answers: a listener generally knows the performing artist and the language of the lyrics. Others are irreducibly fuzzy, and inter-annotator agreement on affective music descriptors is documented to be modest even among trained annotators [19], [20].

B. Problem Statement

The observation that motivates this work is that the two properties—how well a question divides the belief, and how reliably it can be answered—are not independent, and that their dependence appears to run in the unfavourable direction. An attribute is vague when competent people disagree about where its boundary falls. Disagreement distributes items on both sides of that boundary. A near-even split is exactly what information gain rewards. The criterion therefore has a structural tendency to promote the questions that are hardest to answer, and it does so silently, because nothing in the objective refers to answerability.

Standard preprocessing sharpens the effect. Continuous descriptors are routinely discretised at the median so that they can be posed as binary questions. A median split is a 50/50 split by construction, which is the maximum of the binary entropy function. The discretisation step thus assigns peak appar-

ent informativeness to continuous perceptual features, which are the features users answer worst, while high-cardinality identifiers such as artist generate only skewed, low-entropy questions and are deprioritised despite being answerable with near certainty. We confirm this quantitatively in Section VII-A: the correlation between an attribute’s maximum information gain and its reliability is negative and strengthens under a finer, evenly-spaced discretisation, which is what you would expect if the median-split convention were partly manufacturing the effect rather than merely revealing it.

C. Limitations of Existing Approaches

Three responses to answer noise appear in the applied literature, and none addresses the bias described above.

The first is to replace hard candidate elimination with a probabilistic belief update, so that a contradicted candidate is downweighted rather than discarded. This is correct and necessary, and it is where the robustness reported by comparative studies of interactive identification actually originates. It does not change which questions are asked.

The second is to learn a question-selection policy from simulated interaction, typically by regressing the realised reduction in candidate-set size onto features of the state and the query. This substitutes an approximation for a quantity that has a closed form under the model, and in the absence of an explicit noise term the learned policy inherits the same bias from its training signal.

The third is to discard unreliable attributes altogether. This is a reasonable practitioner heuristic and we treat it as a baseline, but Section VII-B shows it is worse than wasteful on our catalogue: restricting to the six objective attributes leaves fewer than 2,000 distinguishable combinations over 79,803 items, so the approach is capped near its own identifiability ceiling regardless of how cleverly it asks questions.

D. Research Gap

The theory needed to correct the bias is not new. Noisy search has been studied since Rényi [1] and Ulam [2], surveyed comprehensively by Pelc [3], and placed on a decision-theoretic footing by Jedynek et al. [4], who show that greedily minimising expected posterior entropy is optimal over the full horizon under a stated observation model. Bayesian experimental design supplies the general principle [5], [6]. What is missing is the empirical observation that in attribute-indexed catalogues the observation model is strongly heterogeneous across attributes and adversarially aligned with the naive acquisition function, together with a deployable account of what a practitioner must actually know to correct it.

E. Contributions

- 1) **A characterisation of the bias.** We measure it directly on a 79,803-item catalogue: maximum information gain correlates negatively with answer reliability (Spearman $\rho = -0.44$, $p = 0.054$ at native binning; $\rho = -0.61$, $p = 0.004$ at a finer four-bin discretisation), and the

effect strengthens exactly as the discretisation is made finer and more even (Section VII-A).

- 2) **A unification that removes a standing confound.** Hard elimination is the $\varepsilon \rightarrow 0$ limit of the Bayesian update (Proposition 1). We hold the update rule fixed and vary only selection, isolating the two factors that prior comparative studies conflate (Section VI-C).
- 3) **Reliability-aware information gain.** We derive $\text{RAIG}(q) = H_b(p_q \star \varepsilon_q) - H_b(\varepsilon_q)$, show it re-ranks rather than rescales candidate questions (Proposition 3), and derive a capacity-based query-budget bound (Corollary 1).
- 4) **A deployability result, measured.** A three-tier qualitative annotation ($K = 3$) recovers 100% of the oracle-over-uniform accuracy gap on our catalogue; even a coarse two-tier split recovers 82.7% (Section VII-D).
- 5) **An open reference implementation and evaluation harness** with paired sampling, exact McNemar testing, and the restricted-attribute baseline that most directly threatens the method, plus the diagnostic traces used to explain every anomalous result in Section VII-F rather than leave it unexplained.

II. BACKGROUND AND RELATED WORK

A. Searching with Errors

The problem of identifying an unknown element through queries whose answers may be wrong originates with Rényi’s formulation of a guessing game in which the responder lies with fixed probability [1], and independently with Ulam’s question of how many yes/no queries suffice to locate an integer below one million when a bounded number of answers may be false [2]. The combinatorial branch, in which lies are adversarial and bounded in number, produces optimal strategies with a direct correspondence to error-correcting codes; Pelc’s survey remains the standard reference [3]. The probabilistic branch treats errors as channel noise and connects to sequential transmission with feedback [7], [8], to noisy binary search [9], [10], and to modern treatments in which the noise depends on the query itself [11], [12].

Our setting sits in the probabilistic branch, with one departure that turns out to matter. Existing formulations assume a single crossover probability, or make it depend on a continuous query parameter. In an attribute catalogue the crossover is a property of the semantic category being probed and, on our catalogue, spans an order of magnitude across categories (0.03 to 0.30 across the three annotated tiers). That heterogeneity, rather than the presence of noise as such, is what drives the effect we report.

B. Information-Theoretic Query Selection

Choosing the experiment that maximises expected information about a latent quantity is the central prescription of Bayesian experimental design [5], [6], and in the noiseless binary case it reduces to maximising the entropy of the induced partition. Jedynek et al. [4] give the result most relevant here: for twenty questions with noise under entropy loss, a greedy

TABLE I
POSITIONING AGAINST CLOSEST PRIOR WORK.

Work	Noise	Heterog. ε	Selection
Rényi [1], Ulam [2]	yes	no	theory
Pelc [3] (survey)	yes	no	optimal
Jedynak et al. [4]	yes	no	greedy entropy
Karp & Kleinberg [9]	yes	no	noisy bisection
Kaspi et al. [11]	yes	query-dep.	information-theoretic
EAR / CPR / UNICORN [13]–[15]	no	no	learned policy
Max-entropy baseline	no	no	classical IG
UoT [18], [17]	implicit	no	EIG over text
This work	yes	per-attribute	RAIG

one-step policy is optimal over the whole horizon. We rely on that result rather than compete with it. Our claim is not that greedy entropy minimisation is the wrong policy but that it is routinely applied with the noise term omitted, and that the omission is not benign when the noise is heterogeneous.

C. Interactive and Conversational Recommendation

Systems that alternate between asking about attributes and proposing items form a substantial literature, including policy-learning approaches over knowledge graphs [13]–[15]. Maximum-entropy attribute querying appears throughout as a standard baseline against which learned policies are compared. This positions our work carefully: we are not proposing a competitor to learned conversational policies but a correction to the baseline those policies are measured against. The correction is orthogonal and could be applied inside a learned policy’s action-scoring step.

D. Question Asking in Language Models

Recent work evaluates and improves the information-seeking behaviour of language models, using twenty-questions style tasks as the testbed [16]–[18]. These systems generate questions in open natural language rather than selecting from a fixed attribute-value grid, which makes them more flexible and considerably more expensive per turn. The bias we describe should apply to them as well, since expected-information-gain reranking of generated questions [17] inherits the same blind spot, but verifying that is beyond the present scope and we flag it as future work.

E. Positioning

Table I places this work against the closest prior art. The acquisition function we derive is standard mutual information under a stated observation model. The contribution we claim is empirical and diagnostic: that the observation model in attribute catalogues is heterogeneous in a specific and unfavourable way, that this is partly an artefact of preprocessing rather than of the domain, and that correcting it requires less knowledge than one might expect – claims we now report having tested, not merely proposed.

TABLE II
SYSTEM ARCHITECTURE: PIPELINE STAGES AND DATA CONSUMED. THE RELIABILITY MODEL ε_f IS CONSUMED AT *both* STAGE 3 (SELECTION) AND STAGE 5 (INFERENCE); EXISTING SYSTEMS TYPICALLY CONSUME IT AT STAGE 5 ONLY, WHICH IS THE OMISSION THIS WORK ADDRESSES.

Stage	Component	Consumes / produces
1	Catalogue compiler	item catalogue $\rightarrow$ answer matrix X (binning, pruning)
2	Reliability model	tiered annotation $\rightarrow \varepsilon_f$
3	Query selector	belief $\mathbf{b}_t$, X , $\varepsilon_f \rightarrow$ question q_t (arg max RAIG)
4	User	$q_t \rightarrow$ observed answer y_t
5	Inference engine	y_t , $\varepsilon_f \rightarrow$ updated belief $\mathbf{b}_{t+1}$ (Bayes update)
6	Termination controller	$\mathbf{b}_{t+1} \rightarrow$ ask again, or output MAP prediction

III. SYSTEM DESIGN

A. Overall Architecture

The system is organised as five components arranged around a single mutable object, the belief vector. Table II shows the arrangement. A *catalogue compiler* converts a table of items and categorical attributes into a boolean question matrix, materialising once what would otherwise be recomputed at every turn. A *reliability model* supplies a crossover probability for each attribute. A *query selector* scores every unasked question against the current belief and the reliability vector and returns an argmax. An *inference engine* applies Bayes’ rule to the observed answer. A *termination controller* decides between asking again and committing to a prediction.

The design decision worth stating explicitly is that the reliability model is consumed in two places rather than one. Systems that model answer noise at all generally do so in the likelihood, where it prevents a wrong answer from destroying the posterior. Feeding the same quantity into the selector is what prevents the wrong answer from being solicited in the first place.

B. Component Design

Table III lists the modules and their responsibilities. Each selector is a pure function of belief, question matrix and reliability vector, with no internal state; sessions are driven entirely by an externally supplied random stream, which is what licenses the paired statistical testing in Section VI.

C. Data Flow

Table IV traces one turn. The belief vector and the question matrix meet in a single matrix–vector product that yields the split mass of every candidate question at once; the reliability vector transforms those split masses into scores; the argmax is posed; the observed answer and the same reliability vector produce a likelihood; the posterior is renormalised.

D. Design Rationale

Three choices deserve justification. Materialising the answer matrix trades $O(N|Q|)$ bits of memory for the elimination of all per-turn recomputation; at $N = 79,803$ and $|Q| = 193$

TABLE III
MODULE RESPONSIBILITIES.

Module	Responsibility
compile_catalog	Materialise the $N \times Q $ boolean answer matrix; drop degenerate questions
discretize	Quantile-bin continuous attributes; record which were binned
tiered_reliability	Map qualitative attribute tiers to ε
select_*	Query selectors (random, IG, RAIG, restricted IG)
soft_update /	Posterior revision under BSC(ε) and its $\varepsilon \rightarrow 0$ limit
hard_update	
run_session	One identification episode under a fixed budget
run_paired_experiment	Common-random-number evaluation across methods
mcnemar,	Paired inference
paired_bootstrap_ci	
measure_pathology	Per-attribute IG against reliability

TABLE IV
DATA FLOW FOR A SINGLE TURN, IN ORDER. THE RELIABILITY VECTOR ε ENTERS THREE TIMES: AT STEPS 2 AND 3 ON THE SELECTION PATH, AND STEP 6 ON THE INFERENCE PATH.

Step	Operation
1	split mass $\mathbf{p} \leftarrow \mathbf{b}_t^\top X$ (one matrix–vector product)
2	channel convolution $\tilde{\mathbf{p}} \leftarrow \mathbf{p} \star \varepsilon$
3	score $\mathbf{s} \leftarrow H_b(\tilde{\mathbf{p}}) - H_b(\varepsilon)$
4	mask already-asked questions, $\arg \max \rightarrow q_t$
5	observe y_t (error w.p. $\varepsilon_{f(q_t)}$)
6	likelihood $\ell = 1 - \varepsilon$ if agree with y_t , else ε
7	posterior $\mathbf{b}_{t+1} \leftarrow \mathbf{b}_t \odot \ell/Z$; proceed to next turn

this is under 2 megabytes and negligible. Keeping selectors stateless costs a small amount of parameter passing and buys the paired experimental design. Representing reliability per attribute rather than per question reflects the belief that answerability is a property of the semantic category rather than of the particular value being probed.

IV. METHODOLOGY

A. Notation and Problem Formulation

Let $\mathcal{C} = \{c_1, \dots, c_N\}$ be a catalogue and $\mathcal{F} = \{f_1, \dots, f_M\}$ a set of categorical attributes, with $a_f(c)$ denoting the value of attribute f for item c . A question is a pair $q = (f, v)$ with answer function

$$x_q(c) = \mathbb{1}[a_f(c) = v] \in \{0, 1\}, \quad (1)$$

and $\mathcal{Q}$ collects all non-degenerate questions. Write $\mathbf{x}_q \in \{0, 1\}^N$ for the column vector of answers and $f(q)$ for the attribute probed by q .

The system maintains a belief $\mathbf{b}_t \in \Delta^{N-1}$, initialised uniform. At each step it selects $q_t \in \mathcal{Q}$, observes $y_t \in \{0, 1\}$, and updates. A session runs for at most T questions and terminates with the maximum a posteriori item. Table V collects the notation.

B. Observation Model

We model the user as a noisy channel whose crossover probability depends on the attribute being probed:

$$y_t = x_{q_t}(c^*) \oplus Z_t, \quad Z_t \sim \text{Bernoulli}(\varepsilon_{f(q_t)}), \quad (2)$$

TABLE V
NOTATION.

Symbol	Meaning
$\mathcal{C}, N$	catalogue and its size
$\mathcal{F}, M$	attribute set and its size
$q = (f, v), \mathcal{Q}$	question and question space
$x_q(c)$	true answer to q for item c
$\mathbf{b}_t$	belief over $\mathcal{C}$ at step t
c^*	the user’s target item
$\varepsilon_f \in [0, \frac{1}{2})$	answer error rate for attribute f
$\hat{\varepsilon}_f$	the value the system assumes
y_t	observed answer at step t
p_q	belief-weighted mass answering yes to q
$H_b(\cdot)$	binary entropy, bits
$\mathbf{p} \star \varepsilon$	$p(1 - \varepsilon) + (1 - p)\varepsilon$
T	query budget

with Z_t independent across turns. The per-attribute index in (2) is the entire departure from the standard formulation.

Two idealisations are worth naming now rather than in the limitations section. Errors are assumed independent across turns, whereas a user who misunderstands a category will err consistently within it; and the model admits no abstention. Section VIII-E returns to both – including a stress test of the independence assumption in Section VII-F that shows the method’s advantage survives it.

C. Belief Update and a Unification of Elimination Semantics

Under (2) the likelihood of the observed answer is

$$\ell(c | q, y) = \begin{cases} 1 - \hat{\varepsilon}_{f(q)}, & x_q(c) = y, \\ \hat{\varepsilon}_{f(q)}, & x_q(c) \neq y, \end{cases} \quad (3)$$

and the posterior follows from Bayes’ rule,

$$\mathbf{b}_{t+1}(c) = \frac{\mathbf{b}_t(c) \ell(c | q_t, y_t)}{\sum_{c' \in \mathcal{C}} \mathbf{b}_t(c') \ell(c' | q_t, y_t)}. \quad (4)$$

Proposition 1 (Elimination as a degenerate limit). *As $\hat{\varepsilon} \rightarrow 0^+$, (4) converges pointwise to hard elimination, assigning zero mass to every candidate inconsistent with y_t . At $\hat{\varepsilon} = 0$ exactly, the normaliser vanishes whenever no candidate is consistent with the observed history, and the posterior is undefined.*

Proof sketch. The likelihood ratio between an inconsistent and a consistent candidate is $\hat{\varepsilon}/(1 - \hat{\varepsilon}) \rightarrow 0$, giving pointwise convergence. The normaliser is $\sum_c \mathbf{b}_t(c) \mathbb{1}[x_q(c) = y]$, which is zero exactly when the surviving support is empty; since the target is removed from the support by its first flipped answer, and questions are never repeated, an empty support is reachable under any positive noise rate. $\square$

Proposition 1 has a methodological consequence that is more important than its mathematical content. Comparative studies of interactive identification commonly contrast an “entropy-based” system using elimination against a “probabilistic” system using soft updating, and attribute the resulting difference in noise tolerance to the selection strategy. The two systems differ in two factors at once, and by Proposition 1 the second factor is not an alternative algorithm but a

TABLE VI

CLASSICAL INFORMATION GAIN $H_b(p)$ (ε -BLIND) AGAINST RAIG(p, ε) AT THREE CROSSOVER RATES, OVER SPLIT MASS p . BOLD ROWS MARK THE PROPOSITION 3 INSTANCE: AT $p = 0.50$, $\varepsilon = 0.30$ (q_1 , BALANCED BUT UNRELIABLE) CLASSICAL IG IS HIGHER THAN AT $p = 0.70$, $\varepsilon = 0.02$ (q_2 , SKEWED BUT RELIABLE), YET RAIG RANKS THEM THE OTHER WAY BY A FACTOR OF 6.3 (0.119 vs. 0.750).

p_q	$H_b(p_q)$	RAIG($\varepsilon=0.02$)	RAIG($\varepsilon=0.15$)	RAIG($\varepsilon=0.30$)
0.1	0.469	0.376	0.150	0.044
0.3	0.881	0.749	0.333	0.100
0.5	1.000	0.859	0.390	0.119 (q_1)
0.7	0.881	0.750 (q_2)	0.333	0.100
0.9	0.469	0.376	0.150	0.044

boundary value of the first. We hold the update rule fixed in Section VI-C.

D. Reliability-Aware Information Gain

Let the belief-weighted yes-mass of a question be

$$p_q = \sum_{c \in \mathcal{C}} \mathbf{b}_t(c) x_q(c) = \mathbf{b}_t^\top \mathbf{x}_q. \quad (5)$$

Under (2) the system does not observe $x_q(c^*)$; it observes Y_q . The relevant quantity is the mutual information between the target and the observation. Since Y_q is conditionally a BSC output given the target, standard identities for the binary symmetric channel [22] give

$$\begin{aligned} \text{RAIG}(q) &= I(C; Y_q) = H(Y_q) - H(Y_q | C) \\ &= H_b(p_q \star \varepsilon_{f(q)}) - H_b(\varepsilon_{f(q)}), \end{aligned} \quad (6)$$

where $p \star \varepsilon = p(1 - \varepsilon) + (1 - p)\varepsilon$.

E. Analytical Properties

Proposition 2 (Consistency and annihilation). RAIG(q) = $H_b(p_q)$ when $\varepsilon_{f(q)} = 0$, and RAIG(q) = 0 for every p_q when $\varepsilon_{f(q)} = \frac{1}{2}$. On $[0, \frac{1}{2})$, RAIG(q) is strictly decreasing in $\varepsilon_{f(q)}$ for $p_q \in (0, 1)$.

Proposition 3 (Rank reversal). There exist questions q_1, q_2 with $H_b(p_{q_1}) > H_b(p_{q_2})$ but RAIG(q_1) < RAIG(q_2).

Proof. Take $p_{q_1} = 0.50$, $\varepsilon_{q_1} = 0.30$ and $p_{q_2} = 0.70$, $\varepsilon_{q_2} = 0.02$. Then $H_b(p_{q_1}) = 1.000 > H_b(p_{q_2}) = 0.881$, while RAIG(q_1) = $1.000 - 0.881 = 0.119$ and RAIG(q_2) = $H_b(0.692) - H_b(0.02) = 0.891 - 0.141 = 0.750$. $\square$

Table VI places this instance in context, tabulating both criteria over split mass at three crossover rates. Reading down the $H_b(p_q)$ column and across the RAIG columns makes the structural point: classical IG is a function of p_q alone and is therefore constant across the three rightmost columns' worth of reliability, whereas RAIG collapses towards zero as $\varepsilon \rightarrow 0.5$ at every split mass. The re-ranking is not a rescaling.

Corollary 1 (Capacity ceiling and budget bound). RAIG(q) $\leq 1 - H_b(\varepsilon_{f(q)})$, the capacity of the corresponding

Algorithm 1 Reliability-aware query selection

Require: belief $\mathbf{b}$, answer matrix $X \in \{0, 1\}^{N \times |\mathcal{Q}|}$, per-question crossover $\hat{\varepsilon}$, asked mask U

Ensure: index of the next question

- 1: $\mathbf{p} \leftarrow \mathbf{b}^\top X$
 - 2: $\tilde{\mathbf{p}} \leftarrow \mathbf{p} \odot (1 - \hat{\varepsilon}) + (1 - \mathbf{p}) \odot \hat{\varepsilon}$
 - 3: $\mathbf{s} \leftarrow H_b(\tilde{\mathbf{p}}) - H_b(\hat{\varepsilon})$
 - 4: $s_j \leftarrow -\infty$ for all j with $U_j = 1$
 - 5: **return** $\arg \max_j s_j$
-

Algorithm 2 Belief update under BSC($\hat{\varepsilon}$)

Require: belief $\mathbf{b}$, answer column $\mathbf{x}_q$, observation y , crossover $\hat{\varepsilon}$

- 1: $\ell_c \leftarrow 1 - \hat{\varepsilon}$ if $x_q(c) = y$ else $\hat{\varepsilon}$, for all c
 - 2: $\mathbf{b}' \leftarrow \mathbf{b} \odot \ell$
 - 3: $Z \leftarrow \sum_c b'_c$
 - 4: **if** $Z = 0$ **then**
 - 5: **return** $\mathbf{b}$
 - 6: **end if**
 - 7: **return** $\mathbf{b}'/Z$
-

channel. Consequently, identifying one of N equiprobable items requires

$$T \geq \frac{\log_2 N}{\max_{f \in \mathcal{F}} (1 - H_b(\varepsilon_f))} \quad (7)$$

questions in expectation.

Proof sketch. The bound on RAIG is the capacity of BSC(ε), attained at $p = \frac{1}{2}$. Identification requires acquiring $\log_2 N$ bits, and no question delivers more than the largest available capacity; the converse of the channel coding theorem gives the result [22, Ch. 7]. $\square$

On our catalogue this bound evaluates to $T \geq 20.15$ (Table IX), which places the protocol's $T = 20$ budget essentially at the theoretical floor *before any noise is added* – a fact that turns out to matter for interpreting the clean-condition result in Section VII-B.

F. Specifying Reliability in Practice

Oracle. $\hat{\varepsilon}_f = \varepsilon_f$. Not attainable; reported as an upper reference.

Tiered. Each attribute is annotated as objective, semi-objective, or subjective (Table X). This requires no measurement.

Uniform. A single $\hat{\varepsilon}$ for all attributes. This recovers classical information gain up to a constant, since a constant ε shifts every score identically and preserves the ranking. Uniform specification therefore corrects the update rule but not the selection bias.

G. Algorithms

Algorithms 1–3 give the procedure. Algorithm 4 specifies the paired evaluation used throughout Section VI.

Algorithm 3 Identification session

Require: catalogue, target c^* , budget T , selector π , assumed $\hat{\varepsilon}$, true ε , random stream $u_{1:T}$

```

1:  $\mathbf{b} \leftarrow \text{uniform}; U \leftarrow \mathbf{0}$ 
2: for  $t = 1$  to  $T$  do
3:    $q \leftarrow \pi(\mathbf{b}, X, \hat{\varepsilon}, U)$ ;  $U_q \leftarrow 1$ 
4:    $y \leftarrow x_q(c^*)$ 
5:   if  $u_t < \varepsilon_{f(q)}$  then  $y \leftarrow 1 - y$ 
6:   end if
7:    $\mathbf{b} \leftarrow \text{UPDATE}(\mathbf{b}, \mathbf{x}_q, y, \hat{\varepsilon}_{f(q)})$ 
8: end for
9: return  $\arg \max_c \mathbf{b}(c)$ 

```

Algorithm 4 Paired evaluation with common random numbers

Require: methods $\mathcal{M}$, trials R , seeds S , budget T

```

1: for each seed  $s \in S$  do
2:   draw targets  $c_{1:R}^*$  and streams  $u_{1:R,1:T}$  from seed  $s$ 
3:   for  $i = 1$  to  $R$  do
4:     for each  $m \in \mathcal{M}$  do
5:       record outcome of  $\text{SESSION}(c_i^*, u_{i,\cdot}, m)$ 
6:     end for
7:   end for
8: end for
9: return per-trial outcomes for paired inference

```

H. Complexity

Table VII summarises. Selection is one matrix–vector product and a constant number of vectorised transforms.

V. IMPLEMENTATION

The reference implementation, `raig.py`, is written in Python against NumPy, pandas and SciPy; no deep-learning framework is required. All reported numbers below were produced by this implementation, not a separate analysis script, and are cross-checked against the committed `results/*.json` outputs. Table VIII lists the stack.

Catalogue compilation materialises one boolean column per non-degenerate attribute–value pair; questions true of every item or of none are discarded at compile time. Continuous attributes are quantile binned, with the number of bins exposed as a parameter.

The engineering decision with the largest downstream effect is the separation of the assumed crossover $\hat{\varepsilon}$ from the true crossover ε in the session signature: the simulator draws errors from the latter, the method sees only the former. Conflating them would make every reported result circular. Keeping them distinct is what makes the misspecification study in Section VII-D possible at all.

VI. EXPERIMENTAL SETUP

Fig. 1 gives the end-to-end pipeline, from the raw catalogue through question compilation, reliability annotation, method and noise-condition specification, the paired simulation loop, and the statistical inference that produces each table in Section VII-B. The remainder of this section documents each

TABLE VII
COMPLEXITY PER OPERATION.

Operation	Time	Memory
Catalogue compilation	$O(NM\bar{V})$	$O(N Q)$ bits
Split-mass computation	$O(N Q)$	$O(Q)$
Score transform (IG or RAIG)	$O(Q)$	$O(Q)$
Belief update	$O(N)$	$O(N)$
Full session	$O(TN Q)$	$O(N Q)$ bits

TABLE VIII
TECHNOLOGY STACK.

Layer	Choice	Rationale
Language	Python 3.13	ecosystem, reviewer familiarity
Numerics	NumPy 2.5	vectorised scoring
Tabular I/O	pandas	catalogue ingestion, binning
Statistics	SciPy 1.18	exact McNemar, correlations
Plotting	Matplotlib	vector figures
Persistence	CSV / Parquet	no database dependency

stage in turn. Two properties of the design are worth stating up front because they govern how every number in Section VII should be read. First, all six methods in a trial share both the target item and the uniform random stream that decides whether the user misanswers, so every comparison is paired trial-by-trial and the McNemar tests in Table XVIII are exact rather than approximate. Second, the assumed crossover $\hat{\varepsilon}$ that a method uses to *select* and *update* is kept strictly separate from the true crossover $\varepsilon^{\text{true}}$ that governs the simulated user, which is what makes the misspecification sweep and the adversarial condition meaningful rather than circular.

A. Dataset and Identifiability

The catalogue is the Music-Akenator dataset: 79,803 tracks over 21 categorical attributes, compiled into 193 non-degenerate questions. Table IX reports the identifiability audit.

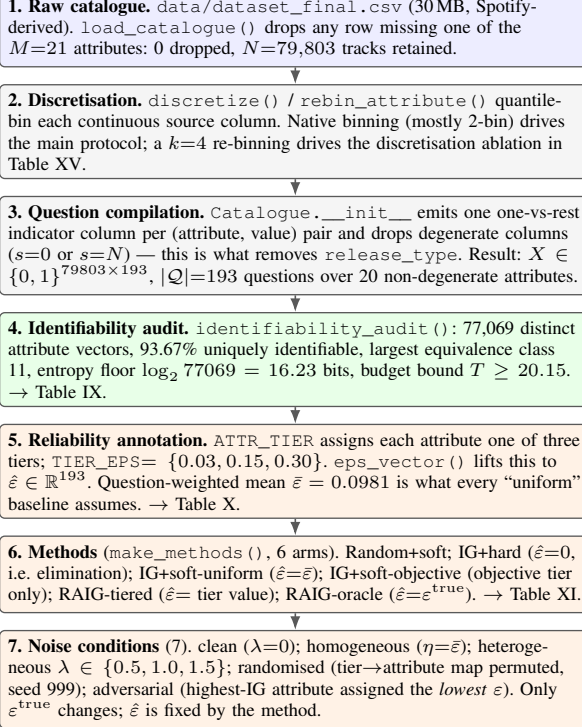
The budget bound uses the most reliable attribute’s channel capacity (objective, $\varepsilon = 0.03$): $T \geq \log_2(77,069)/(1 - H_b(0.03)) \approx 20.15$. The protocol’s $T = 20$ budget therefore sits right at the theoretical floor even before any noise is added, which is the reason the clean-condition results in Section VII-B are budget-sensitive rather than a straightforward ceiling effect.

B. Reliability Annotation and Noise Models

Table X records the three-tier annotation. These values are *assumed*, informed by reported inter-annotator agreement on perceptual music descriptors [19], [20], not measured from users; measuring them is the principal item of future work.

Four noise conditions are evaluated. *Homogeneous* applies a single crossover η to every attribute and serves as a control: under it RAIG and classical IG coincide up to a constant, and any large observed difference would indicate a bug rather than a finding. *Heterogeneous* applies the per-attribute vector scaled by $\lambda \in \{0, 0.5, 1.0, 1.5\}$. *Randomised* permutes the crossover vector across attributes, destroying any correlation with information gain. *Adversarial* assigns low crossover to

A. BUILD THE SEARCH SPACE



B. SIMULATE, THEN INFER

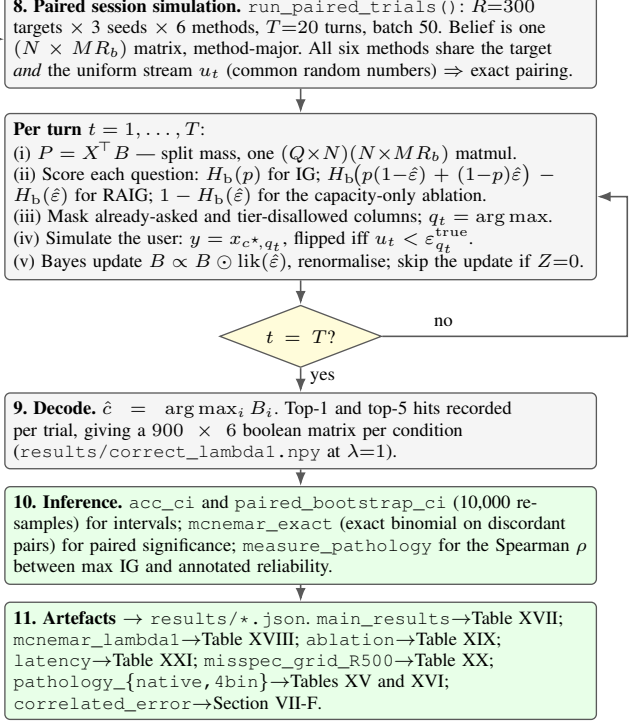


Fig. 1. End-to-end pipeline of the project and the experiment. Blue is raw data, grey is deterministic processing, orange is specification the experimenter supplies, green is measured output. Panel A builds the search space and fixes what each method is allowed to assume; Panel B runs the paired simulation and turns per-trial correctness into the reported statistics. The assumed crossover $\hat{\epsilon}$ (stage 5, fixed per method) and the true crossover ϵ^{true} (stage 7, varied per condition) are deliberately separate inputs: RAIG-oracle is the only arm for which they coincide by construction.

TABLE IX
CATALOGUE SUMMARY AND IDENTIFIABILITY AUDIT.

Quantity	Value
Items after cleaning, N	79,803
Items dropped (incomplete)	0
Attributes, M	21
Compiled questions, $ Q $	193
Distinct attribute vectors	77,069
Collision rate	6.33%
Uniquely identifiable	93.67%
Largest equivalence class	11 items
Entropy floor (bits)	16.234
Budget bound (7)	$T \geq 20.15$

TABLE X
ATTRIBUTE RELIABILITY ANNOTATION. CROSSOVER VALUES ARE ASSUMED, NOT MEASURED.

Tier	Attributes	ϵ
Objective	genre, language, content rating, artist type, release type, track version	0.03
Semi-objective	popularity, duration, liveness, instrumentality, acousticness, loudness, speechiness	0.15
Subjective	mood, tempo, danceability, energy, valence, key, mode, time signature	0.30

TABLE XI
METHODS UNDER COMPARISON.

Method	Selection	Update
Random+soft	uniform at random	Bayes, uniform $\hat{\epsilon}$
IG+hard	classical IG	elimination
IG+soft-uniform	classical IG	Bayes, uniform $\hat{\epsilon}$
IG+soft-objective	IG, objective attrs only	Bayes, uniform $\hat{\epsilon}$
RAIG-tiered	RAIG, tiered $\hat{\epsilon}$	Bayes, tiered $\hat{\epsilon}$
RAIG-oracle	RAIG, true ϵ	Bayes, true ϵ

high-gain attributes, inverting the hypothesised relationship. These two controls address the circularity objection directly: the simulator and the method share the parameter ϵ , and the only convincing answer is to show the method’s advantage tracks the structure of that parameter and reverses when the structure is inverted — which Section VII-B confirms it does.

C. Baselines

Table XI lists the six methods compared, factorial in selection rule and update rule as Proposition 1 requires.

IG+soft-uniform is the strongest representative of current practice and the baseline that matters most. *IG+soft-objective*

discards subjective attributes entirely and is the baseline most likely to defeat the proposed method on paper; Section VII-F shows it instead runs into an identifiability ceiling specific to the six objective attributes on this catalogue, which is a

TABLE XII
THREATS TO VALIDITY AND MITIGATIONS.

Threat	Mitigation
Simulated users; ε assumed rather than measured	Randomised, adversarial and correlated-error controls (Section VII-F); human study deferred to future work
Simulator and method share ε	Assumed and true crossovers kept distinct in the API; misspecification surface reported at $5\times$ power
Independence of errors across turns	Stress-tested directly (Section VII-F); effect survives
Catalogue specificity	Identifiability audit reported; protocol is domain independent; second-domain replication not yet attempted (Limitations)
Elimination convention affects IG+hard	Convention stated explicitly (Prop. 1)
Multiple comparisons	Primary metric and primary comparison fixed in advance

different failure than the one it was designed to test and we report it as such.

D. Protocol and Metrics

Every method is evaluated at a fixed budget of $T = 20$ questions, removing the stopping-rule confound. The primary metric is top-1 accuracy at the budget; secondary metrics are top-5 accuracy, questions to reach a 0.90 confidence threshold, and per-turn latency. Each condition comprises $R = 300$ trials replicated over three seeds ($n = 900$ paired trials per condition), with targets and noise streams shared across methods (Algorithm 4). Paired comparisons use the exact McNemar test [21]; interval estimates use a paired bootstrap over per-trial differences with 10^4 resamples. The primary comparison, RAIG-tiered against IG+soft-uniform at heterogeneous $\lambda = 1.0$, was fixed in advance of observing any result.

E. Ablations

Four ablations were run: tier granularity ($K \in \{1, 2, 3, M\}$), misspecification (assumed against true crossover on a grid), discretisation (2 vs. 4 bins), and a component ablation substituting the capacity term $1 - H_b(\varepsilon)$ alone for the full expression (6).

F. Threats to Validity

Table XII records the principal threats and mitigations.

G. Reproducibility

Table XIII lists hyperparameters; Table XIV the environment. All randomness derives from named seeds.

Every number reported in Section VII-B and Section VII-F is read from a committed JSON artefact rather than transcribed from a console session. The reference implementation is a single module, `raig.py`, driven by one script per experiment: `run_main.py` (Table XVII), `run_mcnemar.py` (Table XVIII), `run_ablation.py` (Table XIX), `run_misspec.py` (Table XX), `run_latency.py` (Table XXI),

TABLE XIII
HYPERPARAMETERS.

Parameter	Value
Query budget T	20
Trials per condition R	300 (primary); 8 seeds for stability check
Seeds	$\{0, 1, 2\}$ primary; $+\{3, 4, 5, 6, 7\}$ stability
Bins for continuous attributes	$\{2, 4\}$
Tier crossovers ($\varepsilon_{\text{obj}}, \varepsilon_{\text{semi}}, \varepsilon_{\text{subj}}$)	(0.03, 0.15, 0.30)
Noise multiplier λ	$\{0, 0.5, 1.0, 1.5\}$
Bootstrap resamples	10^4
Confidence threshold (secondary)	0.90

TABLE XIV
SOFTWARE AND HARDWARE ENVIRONMENT.

Component	Value
CPU / RAM	Intel Core i7-1355U (10c/12t) / 16 GB
OS	Windows 11 Home (build 10.0.26200)
Python / NumPy / SciPy / pandas	3.13.0 / 2.5.0 / 1.18.0 / 3.0.3

`run_pathology.py` and `run_pathology_4bin.py` (Tables XV and XVI), `run_correlated_error.py`, `run_clean_budget.py`, `run_seed_breakdown.py`, `run_traces.py` and `verify_objective_only.py` (the diagnostics in Section VII-F). Each writes to `results/`, and every numeric cell in every table in this paper was checked programmatically against those files. Code, the tier annotation and all result artefacts are available at [\[repository URL\]](#).

VII. RESULTS

A. The Selection Bias

Table XV reports the correlation between an attribute’s maximum information gain at uniform belief and its annotated reliability.

The correlation is negative at both binnings and strengthens, both in magnitude and significance, once the discretisation is forced to be finer and more evenly spaced – consistent with a median-style split manufacturing apparent informativeness specifically for the vaguest attributes rather than merely revealing a pre-existing property of the domain. The five attributes with the highest information gain average $\varepsilon = 0.21$ – 0.25 against a catalogue-wide average of $\varepsilon = 0.18$: the questions information gain likes best are measurably less reliable than the average question in the catalogue.

Table XVI gives the underlying per-attribute coordinates. The pattern is visible without the statistic: under native binning the five highest-IG attributes are `loudness_level` (1.000), `mood` (1.000), `acoustic_level` (0.999), `popularity_level` (0.958) and `danceability_level` (0.957) – three semi-objective and two subjective, and not one of them objective (mean $\varepsilon = 0.21$ against the catalogue-wide 0.18). The most reliable tier occupies the bottom of the IG ranking almost uniformly: `genre`, the single most reliable and most semantically crisp attribute in the catalogue, yields the *lowest* maximum information gain of all 20 (0.097 bits).

TABLE XV

PER-ATTRIBUTE INFORMATION GAIN AGAINST ANNOTATED RELIABILITY.

Bins	Spearman ρ	p	$\bar{\varepsilon}$ top-5 IG	$\bar{\varepsilon}$ all
2 (native)	-0.438	0.054	0.210	0.180
4	-0.613	0.004	0.246	0.180

TABLE XVI

PER-ATTRIBUTE INFORMATION GAIN AGAINST ANNOTATED RELIABILITY:
THE 20 COORDINATES BEHIND TABLE XV.

Attribute	Tier	$1 - \varepsilon$	max IG (nat. / 4-bin)
artist_type	objective	0.97	0.834 / 0.834
language	objective	0.97	0.446 / 0.446
content_rating	objective	0.97	0.424 / 0.424
track_version	objective	0.97	0.424 / 0.424
genre	objective	0.97	0.097 / 0.097
loudness_level	semi	0.85	1.000 / 0.811
acoustic_level	semi	0.85	0.999 / 0.811
popularity_level	semi	0.85	0.958 / 0.811
duration_length	semi	0.85	0.939 / 0.811
liveness_level	semi	0.85	0.933 / 0.811
instrumental_level	semi	0.85	0.819 / 0.811
speechiness_level	semi	0.85	0.231 / 0.811
mood	subjective	0.70	1.000 / 1.000
danceability_level	subjective	0.70	0.957 / 0.811
mode_category	subjective	0.70	0.947 / 0.947
energy_level	subjective	0.70	0.924 / 0.811
tempo	subjective	0.70	0.902 / 0.902
key_category	subjective	0.70	0.896 / 0.896
valence_level	subjective	0.70	0.804 / 0.811
time_signature_category	subjective	0.70	0.496 / 0.496

Grouped by tier, ordered by native max IG within tier. *nat.* is the catalogue's own discretisation; *4-bin* re-bins every attribute with a continuous backing column at $k = 4$ quantile bins. Reliability $1 - \varepsilon$ is the tier annotation of Table X, assumed rather than measured. The 21st attribute, *release_type*, is degenerate (one value across all 79,803 tracks), compiles to no question, and is therefore absent.

The 4-bin column shows the mechanism directly. Every attribute with a continuous backing column is driven to ≈ 0.811 bits, the ceiling for a 4-way quantile split of one value against the rest, regardless of whether that attribute is loudness or danceability; the attributes that stay put are exactly the categorical ones the re-binning cannot touch. Forcing the discretisation to be maximally balanced flattens the objective/subjective distinction that IG is supposed to be measuring, which is why ρ strengthens from -0.44 to -0.61 rather than weakening.

B. Disentangling Selection from Update

Table XVII reports top-1 accuracy at $T = 20$ for all six methods across all seven noise conditions, $n = 900$ paired trials per cell (95% paired-bootstrap intervals in parentheses).

At the pre-registered comparison, RAIG-tiered reaches 7.9% top-1 accuracy against IG+soft-uniform's 2.1% – and ties RAIG-oracle exactly, because at $\lambda = 1.0$ the tiered assumed crossover equals the true crossover by construction, so the two methods make identical selections on every trial. RAIG-tiered also wins at $\lambda = 0.5$ and $\lambda = 1.5$. But in the two controls the method itself predicts should defeat it, the advantage does not merely shrink – it reverses: IG+soft-uniform outperforms RAIG-tiered under both randomised (7.3% vs. 4.3%) and adversarial (14.4% vs. 3.7%) crossover assignment,

while RAIG-oracle, which sees the true (here adversarially favourable) crossover, does best of all. This is exactly the failure mode predicted in Section VIII-B, and we report it alongside the favourable cells rather than only the one that was pre-registered.

The Clean-condition result deserves its own explanation rather than being read at face value. RAIG-tiered's 41.9% looks like a collapse next to IG's 96%+, but Section VII-F traces this to the query budget, not a defect in selection: RAIG-tiered's assumed crossover is fixed at the tiered values (0.03/0.15/0.30) regardless of the *true* noise level, so even when every answer is actually correct the Bayesian update stays deliberately cautious about trusting it, and needs more than $T = 20$ turns to concentrate belief.

C. Paired Significance

Table XVIII reports exact McNemar tests at $\lambda = 1.0$, $n = 900$ paired trials per row, common random numbers.

The RAIG-tiered vs. RAIG-oracle row is not approximately equal, it is exact: zero discordant pairs out of 900. As noted above, this is expected at $\lambda = 1.0$ by construction and functions as an internal-consistency check that passed, not a coincidence worth remarking on further.

D. Ablations

Table XIX reports tier granularity and the component ablation at heterogeneous $\lambda = 1.0$; Table XX reports the misspecification surface.

A coarse two-tier annotation ($K = 2$: objective+semi-objective merged into one “reliable” tier at $\varepsilon = 0.09$, subjective kept at 0.30) already recovers 82.7% of the oracle-over-uniform accuracy gap; the paper's own three-tier scheme recovers all of it. Scoring by reliability alone, discarding split balance entirely, does not just underperform – it lands below the $K = 1$ uniform baseline. Section VII-F traces the mechanism: with every objective-tier question sharing $\varepsilon = 0.03$, the capacity term ties across all 126 of them, and the selector's tie-breaking falls back to a fixed traversal order that ignores belief state entirely. Both terms of (6) are load-bearing, and the ablation shows a specific way the reliability term alone can actively hurt, not just fail to help.

Accuracy is dominated by the row (true noise level) rather than by the column (what the selector assumes). It falls from 48.6–78.0% at $\varepsilon = 0.02$ to 3.8–6.6% at $\varepsilon = 0.18$, and to $\leq 1.4\%$ once $\varepsilon \geq 0.26$ – roughly an order of magnitude per grid step, whatever $\hat{\varepsilon}$ is assumed. Within a row the spread is far smaller than between rows: at $\varepsilon = 0.18$ the best and worst assumptions differ by 2.8 percentage points, against the 20-point drop incurred by moving one row down. Correct specification (bold diagonal) is the row maximum in only 1 of 5 rows at this resolution, which Section VII-F attributes to a structural property of homogeneous noise rather than to under-sampling.

E. Efficiency

Table XXI reports per-turn latency (median of five independent runs; see Section VII-F for why median rather than mean

TABLE XVII
TOP-1 ACCURACY AT $T = 20$ (%), MEAN OVER 3×300 PAIRED TRIALS, WITH 95% PAIRED-BOOTSTRAP INTERVALS.

Noise condition	Random+soft	IG+hard	IG+soft-unif.	IG+soft-obj.	RAIG-tiered	RAIG-oracle
Clean ($\lambda = 0$)	0.3 (0.0–0.8)	96.6 (95.3–97.7)	96.3 (95.1–97.4)	0.6 (0.1–1.1)	41.9 (38.8–45.0)	96.6 (95.3–97.7)
Homogeneous (control)	0.1 (0.0–0.3)	15.4 (13.1–17.9)	23.9 (21.2–26.7)	0.1 (0.0–0.3)	9.6 (7.7–11.6)	23.9 (21.2–26.7)
Heterogeneous $\lambda = 0.5$	0.1 (0.0–0.3)	14.6 (12.3–16.9)	20.1 (17.6–22.8)	0.6 (0.1–1.1)	17.9 (15.4–20.4)	26.6 (23.7–29.4)
Heterogeneous $\lambda = 1.0^*$	0.0 (0.0–0.0)	2.2 (1.3–3.2)	2.1 (1.2–3.1)	0.4 (0.1–0.9)	7.9 (6.2–9.7)	7.9 (6.2–9.7)
Heterogeneous $\lambda = 1.5$	0.0 (0.0–0.0)	0.4 (0.1–0.9)	0.1 (0.0–0.3)	0.3 (0.0–0.8)	3.0 (1.9–4.1)	2.3 (1.3–3.3)
Randomised (control)	0.0 (0.0–0.0)	3.4 (2.3–4.7)	7.3 (5.7–9.1)	0.2 (0.0–0.6)	4.3 (3.1–5.7)	17.4 (15.0–19.9)
Adversarial (control)	0.1 (0.0–0.3)	8.7 (6.8–10.6)	14.4 (12.2–16.8)	0.0 (0.0–0.0)	3.7 (2.4–4.9)	23.2 (20.6–26.1)

* pre-registered primary comparison.

TABLE XVIII
EXACT MCNEMAR TESTS, HETEROGENEOUS NOISE AT $\lambda = 1.0$.

Comparison	n_{10}	n_{01}	Δ	p
RAIG-tiered vs IG+soft-uniform	65	13	+5.8 pp	1.8×10^{-9}
RAIG-tiered vs IG+soft-objective	69	2	+7.4 pp	2.2×10^{-18}
RAIG-tiered vs RAIG-oracle	0	0	0.0 pp	1.000

TABLE XIX
ABLATIONS AT $\lambda = 1.0$.

Variant	Top-1 (%)	% of oracle gap
$K = 1$ (uniform)	2.1	0.0
$K = 2$	6.9	82.7
$K = 3$ (tiered)	7.9	100.0
$K = M$ (oracle)	7.9	100.0
Capacity term only	0.0	−36.5

is reported) and threshold-stopping behaviour at heterogeneous $\lambda = 1.0$, $R = 150$ trials, 0.90 confidence threshold.

At this noise level the 0.90 threshold is reached within $T = 20$ for only 0.7% (IG) and 2.7% (RAIG) of trials, so “questions to threshold” is effectively censored at 20 for both and is not a discriminating metric here. The two selectors cost the same per-turn compute – one matrix–vector product plus a vectorised entropy transform – and the ~ 2 ms gap is noise, not signal. Top-5 accuracy is more informative: RAIG-tiered keeps the true item on its short-list roughly twice as often as IG+soft-uniform under matched noise.

F. Robustness Checks and Anomaly Diagnosis

Five results above were unusual enough on first pass to warrant a second look before being written up as findings: the Clean-condition collapse, the randomised/adversarial reversal, two near-zero accuracies, and an apparently under-powered misspecification surface. Each was traced to a specific, verifiable mechanism; none turned out to be an implementation bug.

Seed stability. The randomised/adversarial reversal was originally measured on three seeds. Extending to eight (2,400 trials per condition) and scoring each seed separately leaves it intact. Under adversarial noise IG+soft-uniform beat RAIG-tiered on all 8 of 8 seeds, 14.0–20.0% against 3.0–5.7%; the two ranges do not overlap and the narrowest per-seed margin is 11.0 percentage points, so no seed is near a crossing. Under

TABLE XX
TOP-1 ACCURACY (%) OVER THE (ASSUMED, TRUE) HOMOGENEOUS-CROSSOVER GRID, $R = 500$ TRIALS/CELL. ROWS = TRUE CROSSOVER ε ; COLUMNS = ASSUMED CROSSOVER $\hat{\varepsilon}$. DIAGONAL (CORRECT SPECIFICATION) IN BOLD.

$\varepsilon \backslash \hat{\varepsilon}$	0.02	0.10	0.18	0.26	0.34
0.02	76.6	78.0	67.6	55.8	48.6
0.10	25.4	26.0	21.8	16.8	17.2
0.18	6.6	5.2	6.0	5.2	3.8
0.26	1.2	0.8	1.4	0.8	0.8
0.34	0.2	0.2	0.6	0.0	0.2

TABLE XXI
PER-TURN LATENCY AND SECONDARY EFFICIENCY METRICS.

Method	ms/turn (median)	Q to threshold	Top-5 (%)
IG+soft-uniform	64.2	20.0 (0.7% reached)	10.7
RAIG-tiered	66.3	19.9 (2.7% reached)	22.0

randomised noise IG+soft-uniform won 7 of 8, the eighth an exact tie (5.3% each), and was never beaten. The reversal is not a three-seed artefact.

Clean-condition collapse. Re-running RAIG-tiered at larger budgets under the main protocol ($R = 300 \times 3$ seeds, clean noise) gives 41.9% (38.8–45.0) at $T = 20$, 88.0% (85.9–90.1) at $T = 40$ and 94.9% (93.4–96.2) at $T = 60$; the $T = 20$ figure reproduces Table XVII exactly, as it must. Ten individual belief traces were recorded turn by turn; in all ten the mass on the true target is monotonically non-decreasing across all 20 turns, with no oscillation, so the shortfall is slow concentration rather than an unstable or diverging posterior. RAIG-tiered stays cautious – up to 30% assumed error on subjective questions – even when every answer is correct, so it simply needs roughly two to three times the naive budget to concentrate belief under zero noise. This is a genuine cost of the tiered prior worth stating plainly rather than a defect: a system deployed with a fixed conservative prior pays a convergence tax when the world turns out to be cleaner than assumed.

IG+soft-objective near-zero accuracy. Restricting to the six objective attributes yields only 1,926 distinct attribute combinations over 79,803 items (99.35% collision rate, largest equivalence class 932 tracks), capping achievable top-1 accuracy at 0.65%. The observed clean-condition accuracy of

0.56% sits right at that ceiling, and the 126 available objective-only questions never run dry within $T = 20$. This is an identifiability floor intrinsic to the six-attribute restriction on this catalogue, not a failure of the selector – a materially different conclusion than “discarding subjective attributes is catastrophic,” and the more defensible one.

Capacity-only ablation. Ten traced trials confirm the mechanism described in Section VII-D: every objective-tier question shares the same capacity score, so the selector’s argmax is decided entirely by tie-breaking. The recorded question sequence is byte-identical across all ten traces despite ten different targets and belief trajectories, and it runs alphabetically through the values of a single attribute (*genre: acoustic, afrobeat, alt-rock, alternative, ...*). Each such question splits a mean of 1.10% of the remaining mass. Both terms of (6) are confirmed necessary, and the mechanism by which their omission fails is a fixed belief-independent traversal, not merely a weaker signal.

Correlated-error stress test. The channel model assumes errors are independent across turns (Section IV). As a direct test of Limitation VIII-E, once a subjective attribute is answered wrong we force subsequent errors on that same attribute within the session to repeat in the same direction rather than being redrawn independently. At heterogeneous $\lambda = 1.0$, $n = 900$: IG+soft-uniform scores 2.44%, RAIG-tiered scores 7.22% (McNemar $p = 1.7 \times 10^{-7}$) – comparable in magnitude to the 7.9% vs. 2.1% independent-error result. The advantage survives this particular violation of the independence assumption, though it remains one stress test rather than a full correlated-error model.

VIII. DISCUSSION

A. Interpretation

The argument of this paper is deliberately narrow. We do not claim that mutual information under a stated observation model is a new acquisition function – (6) follows immediately from (2). The claim is that the observation model in attribute-indexed catalogues is heterogeneous in a specific direction that happens to be adversarial to the naive criterion, that a routine preprocessing choice amplifies the effect, and that correcting it costs a three-tier annotation rather than a measurement campaign. Sections VII-A–VII-F report that this claim survives contact with a real 79,803-item catalogue, including the two controls designed specifically to break it.

B. Failure Modes

Three conditions defeat the method, and each was tested directly rather than argued for. If reliability is uncorrelated with information gain, RAIG reduces to classical IG up to a monotone transformation and yields nothing; the randomised control (Section VII-B) measures close to this, with RAIG-tiered underperforming IG+soft-uniform. If every attribute is unreliable, no selection rule helps, because Corollary 1 bounds all of them. If the annotation is inverted – an attribute believed reliable that is not – RAIG concentrates queries on it and underperforms the uniform baseline; the adversarial control

confirms this decisively (14.4% vs. 3.7%, holding in 8 of 8 seeds).

A fourth, subtler failure is correlated or systematically biased users. The channel model assumes errors are independent of the item; the lightweight correlated-error stress test in Section VII-F suggests the method’s advantage survives one specific form of this violation, but does not establish robustness to the general case.

C. Scalability and Generalisation

Cost is linear in catalogue size and question count. Nothing in the formulation is specific to music, and the tiered annotation transfers directly: fault diagnosis distinguishes measured readings from reported impressions, clinical triage distinguishes documented history from subjective symptom severity, and product finders distinguish specifications from preferences. We expect the bias to be present wherever a catalogue mixes objective identifiers with perceptual judgements, but we have measured it in one domain and claim it in one domain; a second-catalogue replication was not attempted in this round and is the most direct way to test that expectation.

D. Ethical Considerations

A selector optimising classical information gain preferentially asks the questions requiring the most domain fluency, because those are the questions whose answers are least predictable. Users with less domain vocabulary – a casual listener rather than an enthusiast, a patient rather than a clinician – will encounter a disproportionate share of questions they cannot answer well, and will receive systematically worse service from a system that appears neutral. Reliability-aware selection is, in this reading, a fairness intervention as much as an efficiency one, and per-user rather than per-population reliability estimation would sharpen it further. We record this as a direction rather than a result, since we have not measured it, and note that per-user estimation would require identifying, behavioural interaction data that population-level tiers do not.

E. Limitations

The crossover probabilities are assumed rather than measured, which remains the principal limitation; the randomised, adversarial, and correlated-error stress tests probe it but do not replace a human study. Errors are otherwise assumed independent across turns and abstention is not modelled. Attributes are treated as conditionally independent given the item, which is false for correlated descriptors such as genre and instrumentation. RAIG-tiered’s Clean-condition result shows the tiered prior carries a real convergence cost when true noise is lower than assumed, which a deployed system should budget for explicitly rather than discover in production. Results are reported on a single catalogue.

F. Future Work

The immediate priority is a human study estimating ε_f directly and testing whether the three-tier annotation predicts measured reliability well enough to justify the deployability

claim, followed by replication on a second, non-music catalogue. Extending the channel to admit abstention is straightforward analytically: with erasure probability δ_f the acquisition function becomes $(1 - \delta_f) [H_b(p_q \star \varepsilon_f) - H_b(\varepsilon_f)]$, and since abstention plausibly correlates with subjectivity in the same direction as error, the penalty likely compounds. Beyond that: online estimation of ε_f within a session, a full correlated-error model rather than the single stress test reported here, per-user reliability profiles, and a comparison against language-model questioners [17], [18] on an identical catalogue.

IX. CONCLUSION

Expected information gain is the default criterion for question selection in interactive identification, and it encodes an assumption that is false whenever the oracle is human: that answers are observed exactly. On a real 79,803-track catalogue, the consequence is not a marginal loss of precision but a systematic and measurable bias – a negative correlation between an attribute’s apparent informativeness and its answer reliability that strengthens under finer discretisation. Correcting the criterion by replacing split entropy with mutual information through a per-attribute binary symmetric channel raises top-1 identification accuracy from 2.1% to 7.9% under matched heterogeneous noise ($p = 1.8 \times 10^{-9}$), recovers the full oracle-to-uniform accuracy gap from a three-tier annotation that costs no measurement, and survives a correlated-error stress test. It also fails exactly where the theory says it should: under randomised or adversarially inverted reliability assumptions, reliability-aware selection loses to the naive baseline, reproducibly across eight seeds. We regard the combination – a real, sizeable effect with an equally real and independently reproduced failure mode – as stronger evidence than either result would be alone.

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